MATH3023 Advanced Mathematics Applications

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Contents

Preface

THESE NOTES follow the course *Advanced Mathematics Applications*, taught by Lyle Noakes at the University of Western Australia in Semester 2 of 2026. The unit teaches advanced techniques of calculus, including multivariable calculus, Green's, Stoke's and divergence theorem, complex analysis, ordinary and partial differential equations, and Sturm-Liouville theory.

0.1 More detailed overview of the subject.

THESE NOTES are about something but with more detail.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every *or* for all
$\exists$	There exists
$\in$	In *or* is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not *or* without
$\varnothing$	Empty set
$\sim$	Relates to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$\lvert n \rvert$	The absolute value of n
$\square$	It has been proven

Chapter 1

Review of Multivariable Calculus

1.1 Lecture 1

1.1.1 Double and Triple Integrals

To INTEGRATE a scalar function $z = f(x,y)$ over a bounded region, we can take either horizontal or vertical slices, and sum them. For example, to calculate $\iint (f(x,y) = xy)\, dA$ over the region bound by $y = \sqrt{x}$ and $y = x^3$ (fig. 1.1).

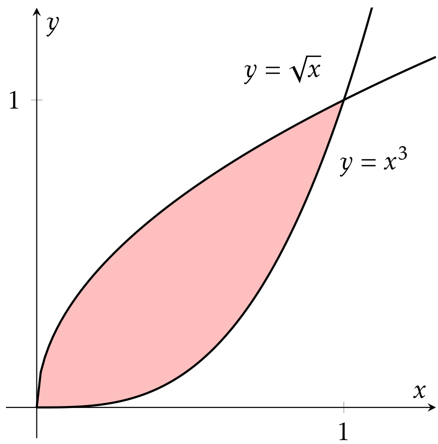

Figure 1.1: The region bound by $y = \sqrt{x}$ and $y = x^3$.

We can either take vertical slices (i.e take regular portions of x and integrate w.r.t y first),

$$\int_0^1 \int_{x^3}^{\sqrt{x}} (xy)\, dy\, dx = \int_0^1 \left[\frac{xy^2}{2}\right]_{x^3}^{\sqrt{x}} dx$$

$$= \frac{1}{2}\int_0^1 x^2 - x^7\, dx = \frac{1}{2}\left[\frac{x^3}{3} - \frac{x^8}{8}\right]_0^1$$

$$= \frac{1}{2}\left(\frac{1}{3} - \frac{1}{8}\right) = \frac{5}{48}$$

Or we can take horizontal slices (i.e take regular portions of y and integrate w.r.t x first),

$$\int_0^1 \int_{x=y^2}^{x=y^{1/3}} (xy)\,dx\,dy = \int_0^1 \left[\frac{x^2 y}{2}\right]_{y^2}^{y^{1/3}} dy$$

$$= \frac{1}{2}\int_0^1 y^{5/3} - y^5\,dy = \frac{1}{2}\left[\frac{3y^{8/3}}{8} - \frac{y^6}{6}\right]_0^1$$

$$= \frac{1}{2}\left(\frac{3}{8} - \frac{1}{6}\right) = \frac{5}{48}$$

Both approaches yield the same result, but one may be easier than the other. For example,

$$\int_{y=0}^{y=1} \int_{x=\sqrt{y}}^{x=1} e^{x^3}\,dx\,dy$$

is impossible, but reversing the order of integration makes it doable.

$$\int_{x=0}^{x=1} \int_{y=0}^{y=x^2} e^{x^3}\,dy\,dx \;= \int_0^1 \left[e^{x^3} y\right]_0^{x^2} dx$$

$$\text{let } u = x^3 : \int_0^1 x^2 e^{x^3}\,dx = \frac{1}{3}\int_0^1 e^u\,du$$

$$= \frac{1}{3}\left[e^u\right]_0^1 = \frac{1}{3}(e-1)$$

Triple integrals are solved in much the same way, as a serious of single integrals.

$$\iiint f(x,y,z)\,dV : a \le x \le b,\ q(x) \le y \le w(x),\ g(x,y) \le z \le h(x,y)$$

$$= \int_a^b \left(\int_{q(x)}^{w(x)} \left(\int_{g(x,y)}^{h(x,y)} f(x,y,z)\,dz \right) dy \right) dx$$

To find the volume of a region R, integrate the function $f(x,y,z) = 1$.

1.1.2 Changing Variables in Double and Triple Integrals

To CHANGE from (x,y) to (u,v), you first need to define a transformation $\mathbf{g}(u,v) = (x,y)$. A common transformation is from (x,y) to (r,θ), with r being the distance from the origin, and θ being the angle from an arbitrary axis.

$$\mathbf{g}(r,\theta) = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Then, define the *Jacobian* matrix to be:

$$\mathbf{J} = \frac{\partial \mathbf{g}(u,v)}{\partial(u,v)} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$$

For the polar coordinates transformation,

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$$

Then,

$$\iint f(x,y)\,dx\,dy = \iint f(x(u,v),y(u,v))|\mathbf{J}|\,du\,dy$$

For polar coordinates,

$$|\mathbf{J}| = r\cos^2\theta + r\sin^2\theta = r$$

So,

$$\iint f(x,y)\,dx\,dy = \iint f(r,\theta)r\,dr\,d\theta$$

The process is the same for triple integrals, except that calculating the determinant by hand is slightly more annoying. When

$$x = x(u,v,w),\ y = y(u,v,w),\ z = z(u,v,w)$$

,

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial x}{\partial y} & \frac{\partial x}{\partial y} & \frac{\partial y}{\partial w} \\ \frac{\partial x}{\partial z} & \frac{\partial x}{\partial z} & \frac{\partial z}{\partial w} \end{bmatrix}$$

1.1.3 Polar Coordinates

As discussed above, polar coordinates involves expressing points on the Cartesian plane in terms of r and θ, where r is the distance from the origin, and θ is the angle from an axis, usually the x-axis. Switching from $(x, y) \to (r, \theta)$ is usually beneficial when the problem being solved is symmetric about rotation around some point.

For example, switch from $(x, y) \to (r, \theta)$ to solve $\iint_R \sin(x^2 + y^2)\,dx\,dy$ where R is the region $1 \leq x^2 + y^2 \leq 16$ (fig. 1.2).

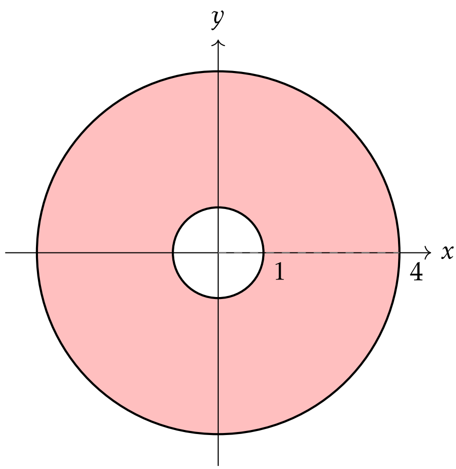

Figure 1.2: Plot of R.

$$\iint_R \sin(x^2 + y^2)\,dx\,dy = \int_0^{2\pi} \int_1^4 r \sin(r^2)\,dr\,d\theta$$

$$\text{let } u = r^2,\ du = 2r\,dr: \int_0^{2\pi}\int_1^4 r\sin(r^2)\,dr\,d\theta = \frac{1}{2}\int_0^{2\pi}\int_1^{16} \sin(u)\,du\,d\theta$$

$$= \int_0^{2\pi} [-\cos(u)]_1^{16}\,d\theta = [(\cos(1)-\cos(16))\theta]_0^{2\pi} \approx 9.4$$

1.1.4 Cylindrical Polar Coordinates

Cylindrical polar coordinates involves changing from $(x,y,z) \to (r,\theta,z)$ (fig. 1.3).

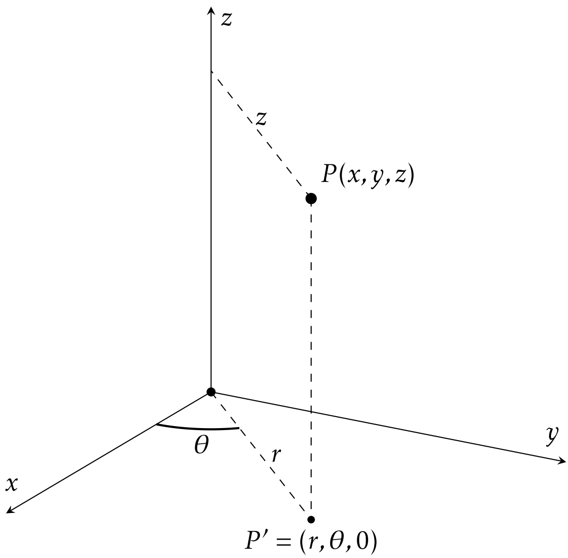

Figure 1.3: Cylindrical Polar Coordinates Projection

The point $P = (x,y,z)$ is projected down to the x,y-plane and then converted to polar coordinates; $P(x,y,0) \to P(r,\theta,0)$. The z-coordinate is then reused. Cylindrical polar coordinates are best used for problems that are symmetric about rotation around the z-axis. Since this is essentially polar coordinates, the determinant of the Jacobian is r.

For example, the mass of a solid cylinder C of radius 1, and height 2, centered at the origin (fig. 1.4) has density

$$D(x, y, z) = 1 + x^2 + y^2 + z^2$$

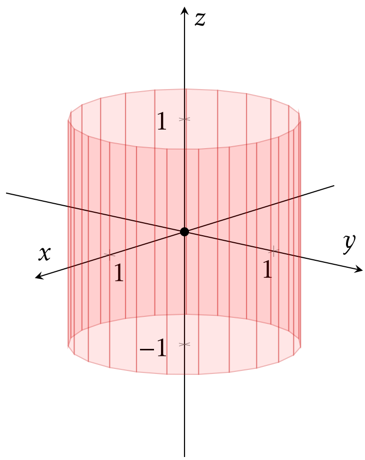

Figure 1.4: Plot of Cylinder C

To find the mass of this cylinder, we need to solve $\iiint_C D(x, y, z)\, dV$

$$\iiint_C D(x, y, z)\, dV = \iiint_{C'} D(r, \theta, z) r\, dr\, d\theta\, dz$$

$$= \int_{-1}^{1} \int_{0}^{2\pi} \int_{0}^{1} r(1 + r^2 + z^2)\, dr\, d\theta\, dz$$

$$= \int_{-1}^{1} \int_{0}^{2\pi} \left[\frac{r^2}{2} + \frac{r^4}{4} + \frac{r^2}{2} z^2 \right]_0^1 d\theta\, dz = \int_{-1}^{1} \int_{0}^{2\pi} \frac{3}{4} + \frac{1}{2} z^2\, d\theta\, dz$$

$$= \int_{-1}^{1} \left[\frac{3}{4}\theta + \frac{1}{2} z^2 \theta \right]_0^{2\pi} dz = \int_{-1}^{1} \frac{6\pi}{4} + \pi z^2\, dz$$

$$= \left[\frac{6\pi}{4} z + \frac{\pi}{3} z^3 \right]_{-1}^{1} = \frac{11\pi}{3}$$

1.2 Lecture 2

1.2.1 Spherical Polar Coordinates

Spherical Polar Coordinates involves changing from $(x, y, z) \to (r, \theta, \phi)$, where r is the distance from the point to the origin, θ is the projected angle from the x-axis (the same as cylindrical polars), and ϕ is the angle from the z-axis to the point projected onto the x, z-plane (taking $y = 0$) (fig. 1.5). The determinant of the Jacobian for spherical polar coordinates is $r^2 \sin\phi$.

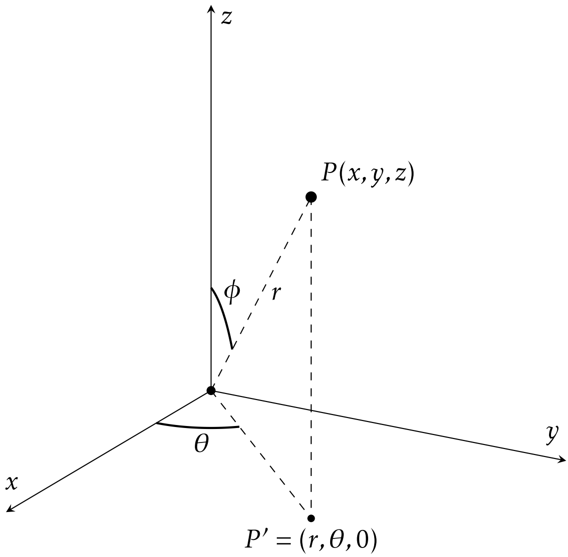

Figure 1.5: Spherical Polar Coordinates

The mass of a sphere of radius 1 centered at the origin with density

$$D(x, y, z) = 1 + x^2 + y^2 + z^2 = 1 + r^2$$

is

$$\int_0^{\pi} \int_0^{2\pi} \int_0^1 (1 + r^2)(r^2 \sin\phi)\, dr\, d\theta\, d\phi$$

$$= \int_0^{\pi} \int_0^{2\pi} \left[\frac{r^3}{3} + \frac{r^5}{5} \right]_0^1 \sin\phi\, d\theta\, d\phi$$

$$= \frac{8}{15}\int_0^{\pi} 2\pi \sin\phi \, d\phi = \frac{16\pi}{15}\left[-\cos\phi\right]_0^{\pi} = \frac{32\pi}{15}$$